BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Operating Systems: Files and Devices

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is an operating-system user interface for?

2. What does a denial-of-service attack try to do?
   A) Improve network speed
   B) Make a service unavailable
   C) Create secure backups
   D) Repair damaged hardware

3. What is data interception?

4. A login uses unsafe input in an SQL query. "SQL injection happens when a database has too many rows." Correct this. How could an attacker affect the data?

Today's objectives

Explain how an operating system manages peripherals and drivers. Describe user accounts, access rights and file-management functions.

Key vocabulary - Read, cover, recall

Peripheral
External or connected input/output hardware.

Device driver
Software enabling the OS to communicate with a particular device.

File management
Creating, naming, saving, organising and moving files/folders.

User management
Creating and controlling user accounts.

BLACK PEN: Write one definition in your own words.

Do Now — **Recall previous learning**
I Do — Read new knowledge
We Do — Practise with guidance
You Do — Work independently
Review & Improve — Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Devices and drivers

The OS manages communication between the CPU and peripherals such as printers. A device driver provides software instructions that allow the OS to communicate with the specific hardware.

User management

The OS creates and manages accounts, passwords and permissions; staff and students may be allocated different access rights.

Files and folders

File management means naming, saving, copying, moving, renaming, deleting and organising files within folders/directories.

Choose the correct service

Device not responding? Consider its driver. A pupil cannot access staff data? Check account rights. Lost a saved document? Locate folders and file paths rather than blaming memory allocation.

I Do - Worked model

A school printer is installed and students can print from a PC. Explain the role of the operating system and a device driver. [4 marks]

- Read the sequence: application -> operating system -> driver -> printer.
- The OS manages communication with peripherals [1] and processes a print request [1].
- A printer driver provides device-specific instructions [1], allowing the computer to communicate correctly with the printer [1].
- OS manages peripherals; driver enables communication with particular hardware.

We Do - Guided practice

1. A college connects a new digital projector to a teacher's PC. Explain why a suitable driver and peripheral management are needed. [4 marks]

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2. A pupil account should not modify the school payroll folder. Explain how the OS helps protect files. [4 marks]

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3. A pupil saves and moves a homework document into a new subject folder. Describe FOUR file-management operations the OS can provide. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the selected menu action.

```
choice = 1
if choice == 1 then
print("Save document")
else
print("Move document")
endif
```

You Do - Independent exam questions

Q2. State the purpose of a device driver. [2 marks]

Q3. A hotel has guest and manager accounts on one shared computer. Explain how user management prevents guests from changing booking records. [4 marks]

Q4. Explain the operating system's roles in file management, device management and user management for a school. [6 marks]

Exit Ticket - True or False?

A device driver helps the operating system communicate with particular hardware.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A user interface lets people use the computer, through GUI controls or typed commands.

2. B - Make a service unavailable

3. Capturing data during transmission, potentially exposing confidential information.

4. Unsafe input can change an SQL query, letting an attacker view or change data. Explain how input changes the query and affects data access or privacy.

I Do - Explain the model

Read the sequence: application -> operating system -> driver -> printer.

The OS manages communication with peripherals [1] and processes a print request [1].

A printer driver provides device-specific instructions [1], allowing the computer to communicate correctly with the printer [1].

OS manages peripherals; driver enables communication with particular hardware.

We Do - Model answers

1. The OS manages communication with the projector [1] as an output peripheral [1]. A suitable driver enables software to send the commands in a format the projector can understand [1], allowing images to be displayed [1].

2. The OS uses user accounts [1] to identify the user [1]. Access rights/permissions limit file and folder access [1], preventing pupil accounts from changing payroll files [1].

3. It allows the pupil to name the file [1], save it to storage [1], create/name a folder [1] and move the document into the folder [1].

Code First - Answer

```
choice = 1
if choice == 1 then
print("Save document")
else
print("Move document")
endif
```

Expected output: Save document

You Do - Self-mark

Q2: Software that enables an operating system to communicate [1] with a specific hardware device/peripheral [1].

Q3: OS assigns different user accounts [1] and can authenticate the person signing in [1]. Access rights restrict guest permissions [1], so only authorised staff can edit booking records [1].

Q4: It names/saves/moves/organises files [1], allowing pupils to locate work [1]. It manages peripherals/drivers [1] so printers and other hardware can communicate [1]. It manages accounts [1] and access rights to restrict confidential files [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.